$\phi : V \longrightarrow \{1, 2, \dots, k\}$

$\phi(u) \neq \phi(v) \quad \forall uv \in E$

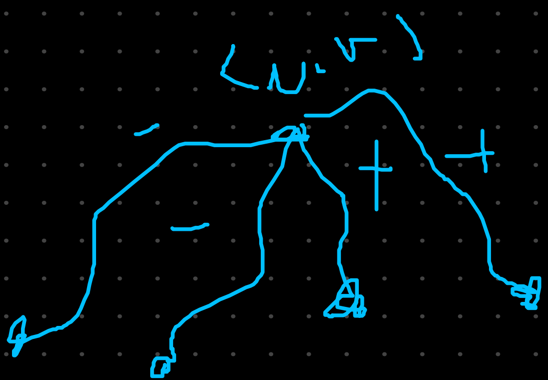
$V = V_1 \sqcup V_2 \sqcup \dots \sqcup V_k$ such that

$V_i = \phi^{-1}(i)$

$VE(G, \phi) = (V, E)$ of

$G = (V, E)$

$v_L(\langle u, v\rangle) \Rightarrow$



— 3-Cycle Problem —

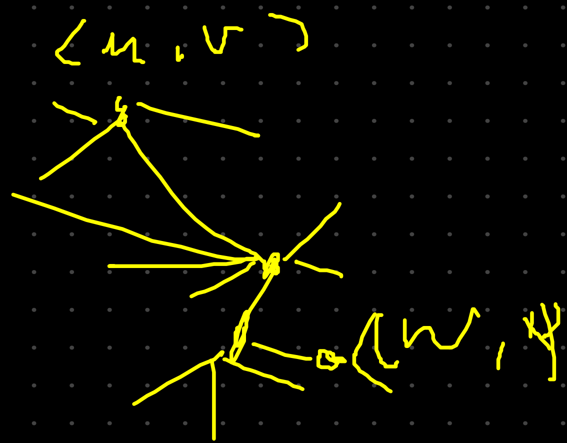
$u, v, w \in V_i$

$u < v < w < u$

$G,\ \mathcal{L} \vee E(h, \phi)$

$\langle e, f\rangle, \langle f, g\rangle, \langle e, g\rangle$

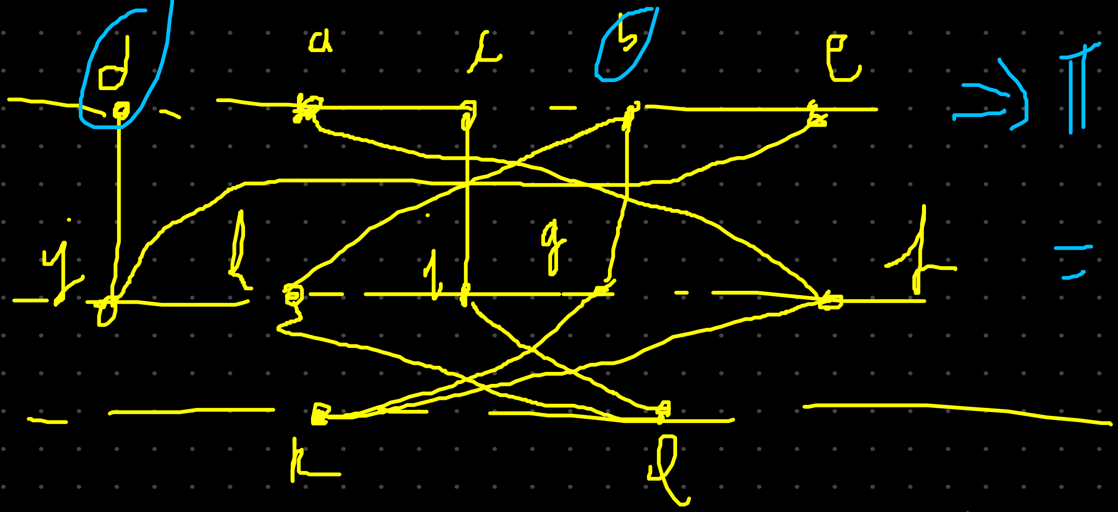
$$\mathcal{L} \xrightarrow[\text{Mapping } \pi]{\text{construct}}$$



$\pi\ V \to V$

$\pi(\langle u, v\rangle) = [v, u]$

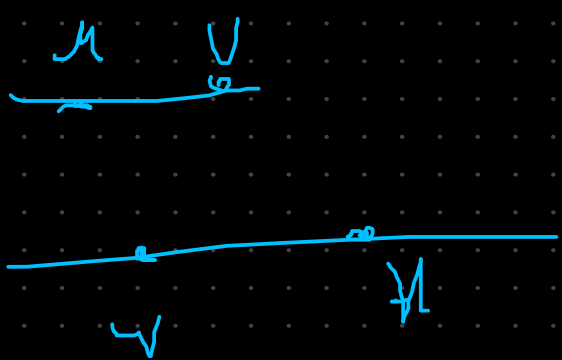
$\pi(\langle w, x\rangle) = [x, w]$

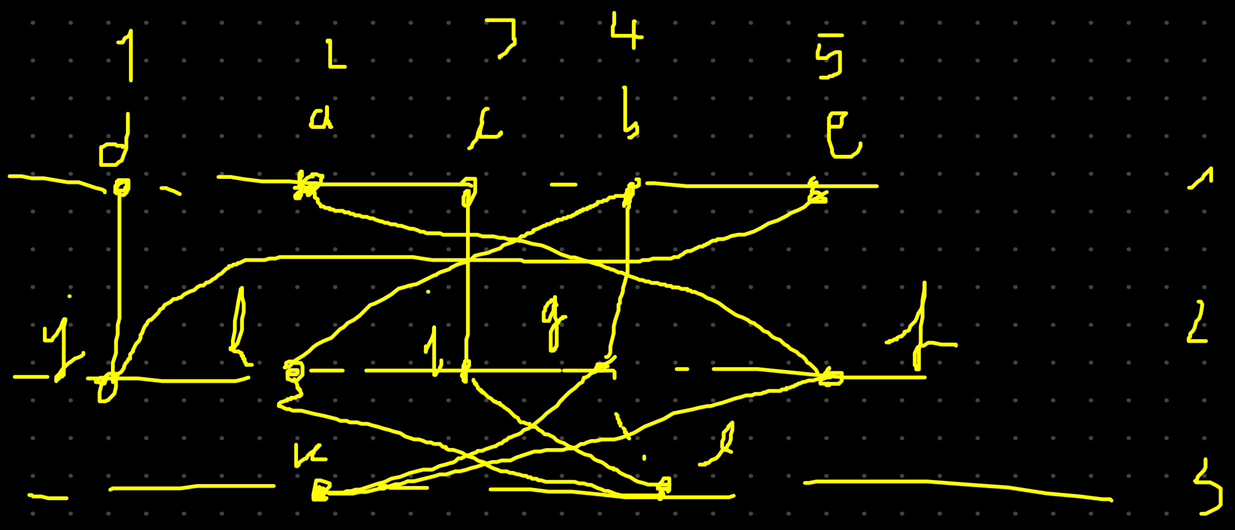


$$\Rightarrow \Pi(\langle d,b\rangle) = [d,b]$$

$$S = (\langle d,b\rangle)$$

$$\Pi(\langle w,x\rangle) \neq \begin{cases} [w,x] & \lambda = + \\ [x,w] & \lambda = - \end{cases}$$

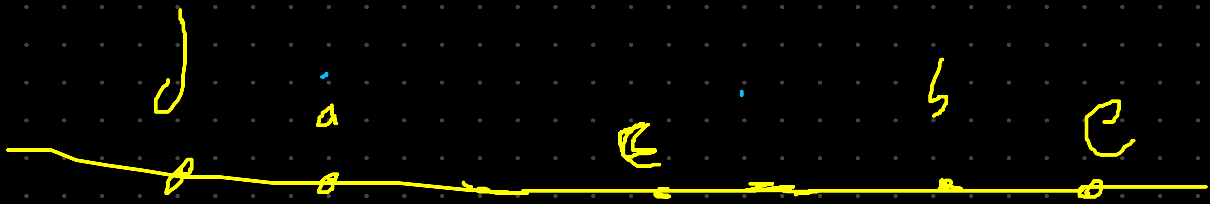


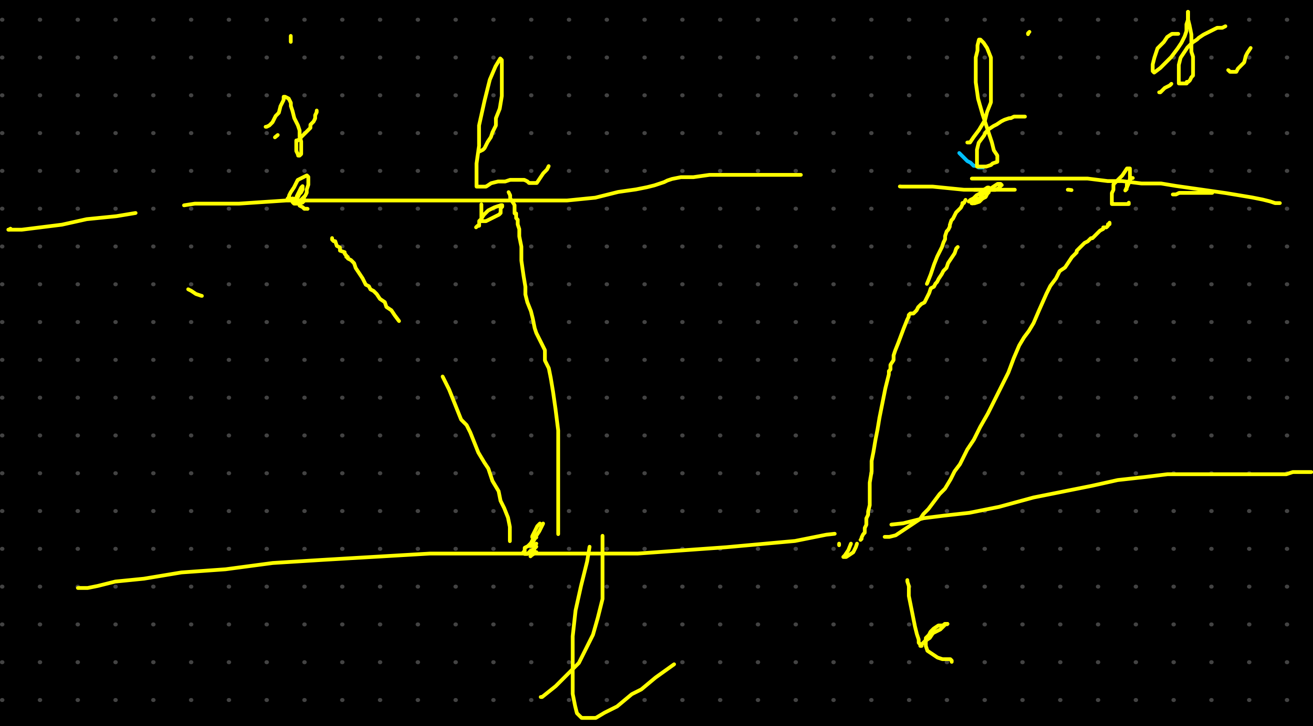


<u, v> = latexit (d/h , i , j)

{i, j} ⇒ | plus cnt = 1

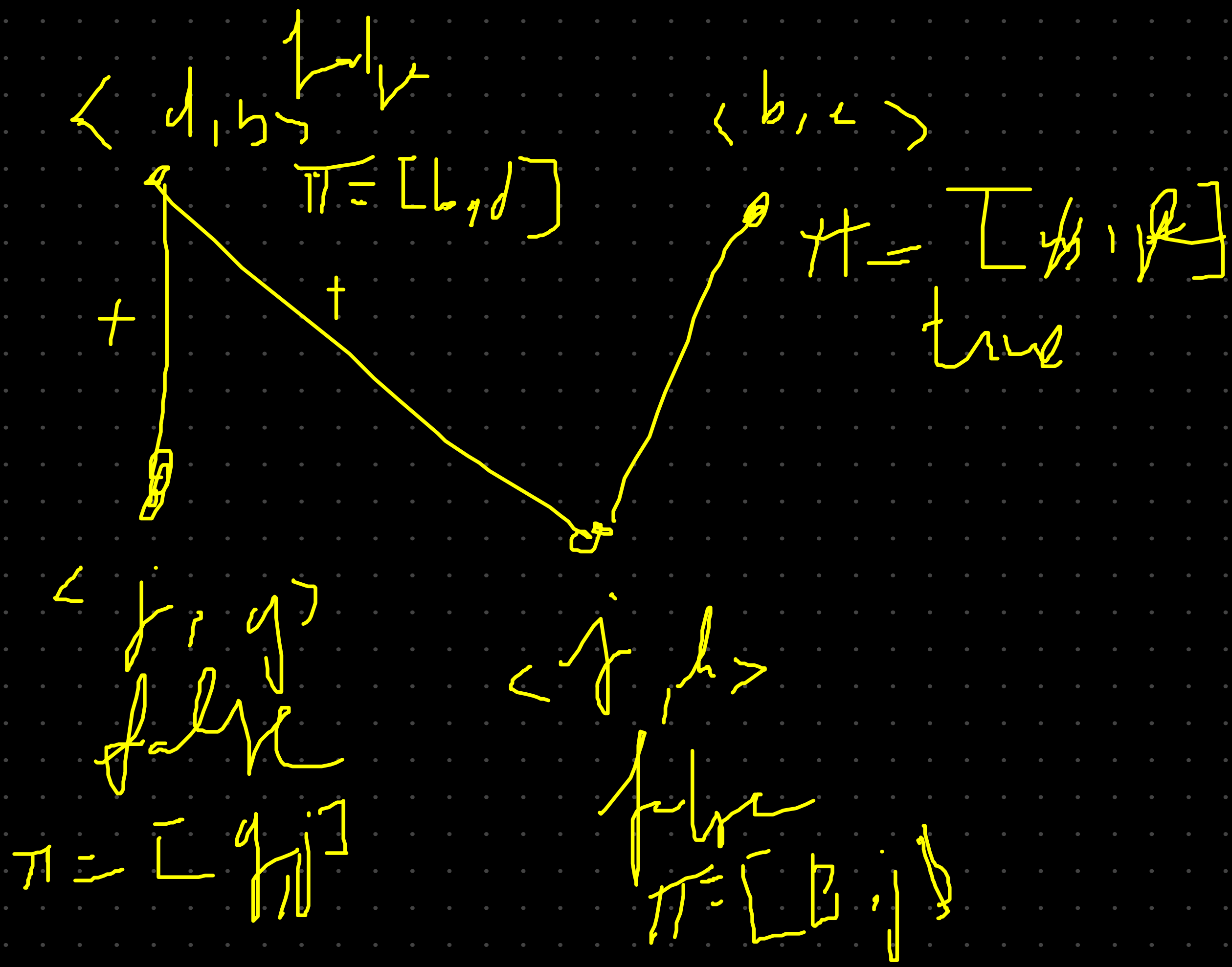
minus cnt = 1





$$S = \left(\langle d, c\rangle, \langle d, e\rangle, \langle a, b\rangle, \langle a, e\rangle, \langle j, g\rangle, \langle b, e\rangle \right)$$

true



<d, b>
true
π = [b, d]
<b, c>
π = [,]
true
t
t
<j, g>
false
π = [g, j]
<j, h>
false
π = [b, j]